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Palaeomagnetism of Precambrian rocks in northern Sweden and its correlation to radiometric data

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Abstract

A palaeomagnetic study has been performed on Palaeo- to Mesoproterozoic basic intrusions and volcanic rocks from the Fennoscandian shield in northern Sweden. Three, possibly four, different generations of magnetizations were identified, the oldest assigned to a Svecofennian age (1.86–1.89 Ga). A second generation is related to the intrusion of granitoids of 1.80–1.76 Ga. In this geological event probably also the third group of directions has its origin. These different magnetizations may indicate that there are at least two different generations of basic intrusions in northern Sweden. A fourth group of directions is isolated as overprints. This magnetization is interpreted to be a Subjotnian magnetization, reflecting a previously unrecognized regional Subjotnian metamorphic event in the northwestern part of the Fennoscandian shield. The drift history for the Fennoscandian shield during the period 1.88–1.50 Ga has been defined based on these new palaeomagnetic data.

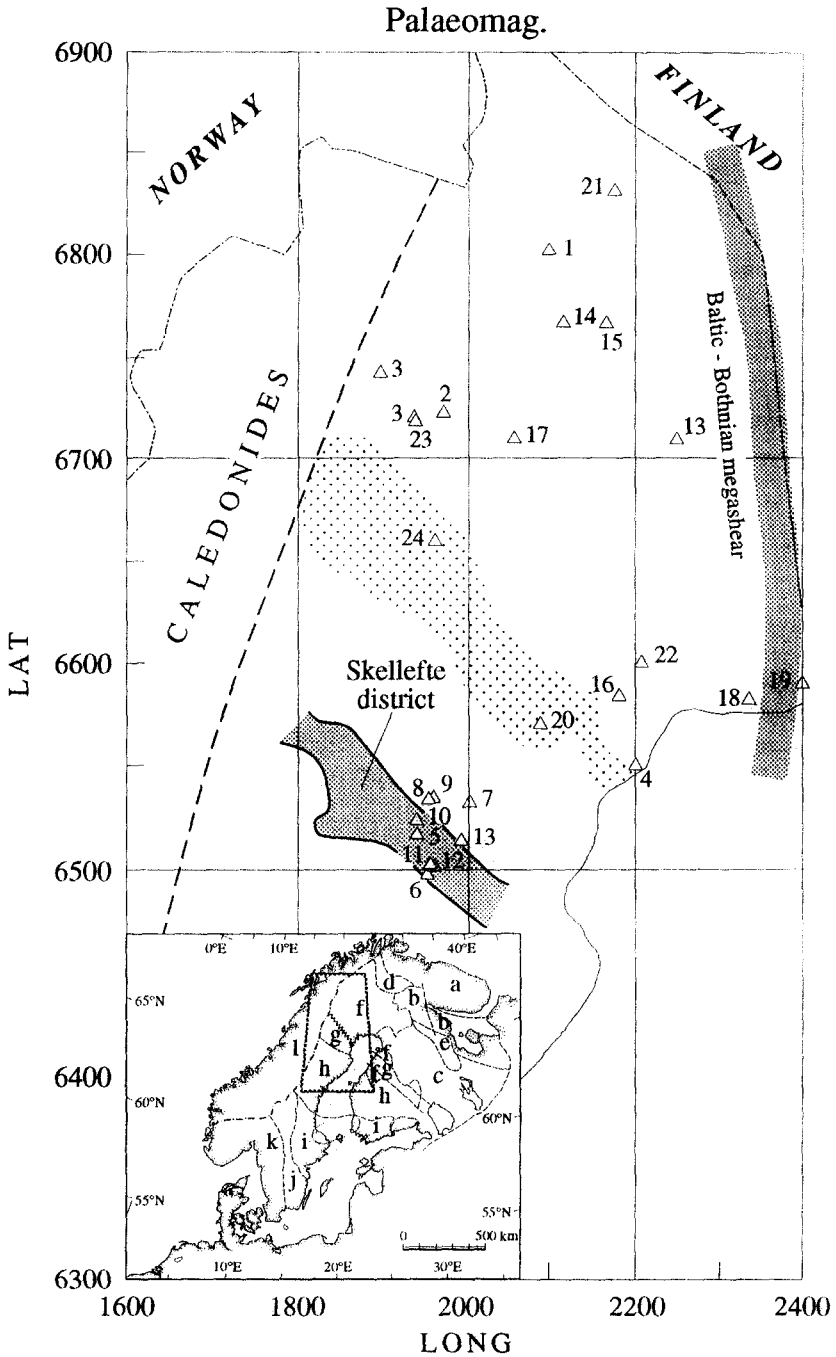
1. Introduction

The Fennoscandian shield may be divided into a number of tectonomagmatic blocks revealing a pattern of geochronological zonation, the oldest parts being in the northeast (Fig. 1). Tectonically, this zonation reflects the growth of the continental crust during a number of successive orogenic events. Palaeomagnetism has proven to be a useful technique for reconstructing the movements of continental blocks before they formed the present-day shields (Irving et al., 1984). However, so far palaeomagnetic studies within the Fennoscandian shield have not demonstrated any intraplate tectonics (Pesonen et al., 1989; Elming et al., 1993).

This study is concentrated on the Skellefte district and the Northern Svecofennian subprovince (the NS-block). Earlier palaeomagnetic studies in this part of the Fennoscandian shield were concentrated on basic intrusive rocks (Piper, 1980; Elming, 1985) and resulted in the development of an apparent polar wander path (APWP) in a time sequence from ~1.9 Ga to 1.5 Ga. Further studies, which included Palaeoproterozoic and Archaean rocks from northern Finland (Mertanen et al., 1989), Sweden and Russia (Elming et al., 1993) resulted in a continuation of the APWP to the Palaeoproterozoic and Archaean.

The purpose of this paper is fourfold. First, to improve the base for a tectonic reconstruction of Fennoscandia during the Palaeoproterozoic. Second, to clarify if the different directions of magnetization reflect different generations of intrusive rocks or if they are the effect of metamorphism. Third, is it possible to identify a Subjotnian geological event in this part of the shield, as

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Palaeomag.
NORWAY
FINLAND
CALEDONIDES
Baltic - Bothnian megashear
Skellefte district
LAT
LONG
6900
6800
6700
6600
6500
6400
6300
1600
1800
2000
2200
2400
0°E
10°E
40°E
65°N
60°N
55°N
10°E
20°E
30°E
0
500 km

was indicated by Piper (1980) and Elming (1985)? Fourth, to look for tectonical differences between the Skellefte block and the NS-block.

2. General geology

Archaean rocks in northern Sweden exist only as relicts as most were destroyed during Proterozoic remobilizations. Skiöld (1987) presented a chronology of late Archaean to Palaeo–Mesoproterozoic episodes of northern Sweden in which the main phase of metamorphism and folding is denoted the Svecofennian (SF) orogeny (1.93–1.85 Ga). The SF province is divided into northern, central and southern subprovinces (Gaál and Gorbatschev, 1987; Fig. 1). Recent deep reflection seismic studies in the Bothnian Bay (BABEL Working Group, 1990) indicate that the volcanic terrain of the Skellefte district, which marks the southern limit of the northern SF subprovince, is probably the result of northeasterly subduction of oceanic crust beneath a crust of continental origin in the north. This subduction led to widespread intrusions of SF granitoids and extrusion of volcanic rocks onto the Archaean craton at 1.89 to 1.86 Ga (Skiöld and Öhlander, 1989). On the basis of the analyses of Sm–Nd isotopes the Archaean palaeoboundary has been defined as closely coinciding with the northern extension of the Skellefte volcanic rocks (Öhlander et al., 1993; Fig. 1).

The emplacement of migmatite-forming granites at 1.80–1.77 Ga was earlier believed to define the end of the SF orogeny but is now regarded as post-SF (Skiöld and Öhlander, 1989). The age of these granites corresponds to the age of the Transscandinavian Igneous Belt (Gorbatschev, 1985; Jarl and Johansson, 1988) in central and southern Sweden. Younger rocks were indicated from Rb–Sr datings (Welin et al., 1971; Gulson, 1972; Skiöld, 1979). These rocks were first believed to define a continuation of the SF orogeny for some 0.25 Ga, but later the low ages were considered to be the result of disturbed radiometric systems and their geological meaning questionable (Skiöld, 1987).

Pb isotope analyses of rocks of the Precambrian basement indicate a Caledonian influence sometimes up to 100 km east of the Caledonian front (Romer, 1990).

3. Geology and age of sampled rocks

Gabbroic rock formations in northern Sweden may from their structure and tectonic style be classified into at least three different categories. One is defined by irregular-shaped massifs (e.g. 13 and 14, Table 2, Fig. 1), a second by well rounded and tectonically undisturbed massifs (e.g. 15 and 16), and a third by layered intrusions (e.g. 1 and 17). Unfortunately very few gabbros have been radiometrically dated. However, the Gallejaur gabbro and a monzonitic part of this massif and the marginal parts of the Vit-

Fig. 1. Location of gabbro massifs and sites of volcanic rocks subject of this palaeomagnetic study. The codes refer to: *1*=Tavinunnanen gabbro, *2*=Akkavaara gabbro, *3*=Nabrejarka gabbro-diabase, *4*=Kallax gabbro, *5*=Gallejaur gabbro, *6*=Bastutjärn gabbro, *7*=Näsberg gabbro, *8*=volcanic rock (Granliden), *9*=volcanic rock (Granliden), *10*=volcanic rock (Stöverberg), *11*=volcanic rock (Örträsk), *12*= volcanic rock (Bjurliden). Codes referring to gabbro massifs of earlier palaeomagnetic studies: *13*=Tärendö, *14*=Svappavaara, *15*=Vittangi, *16*=Notträsk, *17*=Dundret, *18*=Sangis, *19*=Kukkola, *20*=Korsträsk, *21*=Juolovaara, *22*=Niemisel, *23*=Nabrejarka, *24*=Jokkmokk. In the inset the tectonomagmatic block division of northern Fennoscandia is shown (from Elming et al., 1993). The letters and corresponding blocks are as follows: ARCHAEAN province: *a*=Kola, *b*=Belmorian, *c*=Karelian subprovinces; SVECOFENNIAN province: *d*=Lappland granulite belt, *e*=Jatulian/Svecofennian outlier in Karelia, *f*=Northern Svecofennian subprovince, *g*=Skellefte and Raahe Ladoga belts, *h*=Central Svecofennian subprovince, *i*=Southern Svecofennian subprovince, *j*=Transscandinavian Granite and Porphyry Belt; *k*=SOUTHWEST SCANDINAVIAN province; *l*=CALEDONIDES. The Archaean palaeoboundary is marked by the dotted area and the Baltic–Bothnian megashear (Berthelsen and Marker, 1986), one of the main tectonic features within the Northern Svecofennian (NS) subprovince, is marked by the shadowed area. The NS subprovince is in the south limited by the Skellefte district.

Table 1
Palaeomagnetic results from the Nothern Svecofennian subprovince and the Skellefte district

Lat/Long. °N/°E	Locality		N/n	Decl. (°)	Incl. (°)	α_{95}	k	P_{lat}	P_{lon}	D_p/D_m	Magn.
Gabbroic rocks in northern Sweden											
68.03/21.00	Tavinunnanen	(1a)	4/17	2	19	10.4	45	32	199	6/11	SJ (A)
		(1b)	2/6	24	46	–	–	47	169	–	PSF (B)
		(1c)	2/4	6	−10	–	–	17	195	–	SJ (A′)
67.23/19.76	Akkavaara	(2a)	3/10	14	51	6.4	160	54	180	6/9	PSF (B)
		(2b)	3/12	6	15	13.0	39	30	193	7/13	SJ (A)
67.42/19.00	Nabrejarka	(3)	3/24	342	48	1.3	4100	50	224	1/2	SF
65.50/22.00	Kallax	(4a)	5/29	355	42	4.2	226	49	209	3/5	PSF
		(4b)	1/5	21	57	(14.5)	(19)	60	168	–	PSF (B)
65.17/19.43	Gallejaur	(5)	2/12	341	41	–	–	47	225	–	SF
64.97/19.55	Bastujärn	(6)	3/13	343	48	15.8	26	53	224	13/21	SF
65.32/20.05	Näsberg	(7)	3/14	344	35	9.8	69	43	221	7/11	SF
Volcanic rocks in the Skellefte district											
Arvidsjaur group:											
65.37/19.63	Granliden	(8a)	1/2	346	19	–	–	34	216	–	SF (?)
		(8b)	1/4	351	63	(15.5)	(20)	69	217	–	SF
65.37/19.63	Granliden	(9)	1/4	8	28	(12.2)	(33)	39	190	–	SJ (A)
Skellefte group:											
65.25/19.45	Stöverberg	(10)	1/5	4	12	(8.4)	(56)	31	195	–	SJ (A)
65.25/19.45	Örträsk	(11a)	1/3	356	14	(9.9)	(67)	32	204	–	SJ (A)
		(11b)	1/5	341	45	(14.7)	(18)	50	226	–	SF
65.00/19.65	Bjurliden	(12)	1/3	341	20	(15.8)	(26)	34	222	–	SF (?)

Note. Locality = name of the locality and codes for the different directions of magnetization; N = number of sites and n = number of samples; Decl. = mean declination; Incl., = mean inclination; α_{95} = radius of 95% confidence circle; k = Fisher precision parameter, when α_{95} and k are within parentheses they refer to sample means. P_{lat}, P_{lon} = latitude and longitude of calculated palaeomagnetic pole positions; D_p, D_m = semiaxes of 95% confidence oval about the pole, Magn. = symbols for the different generations of magnetization.

tangi gabbro (massifs 5 and 15, Tables 1 and 2) were radiometrically dated as 1.875 Ga (U–Pb, Skiöld et al., 1993) and 1.86 Ga (U–Pb, Skiöld, 1988), respectively.

This study includes samples from seven different gabbro massifs (1–7, Fig. 1) and five sites of volcanic rocks (8–12) located between 64°N and 68°N latitude and 18°E and 24°E longitude.

Two of the most northern gabbro massifs (Akkavaara (2) and Nabrejarka (3)) intrude a perthite monzonite with a Rb–Sr age of 1.53 Ga (Gulson, 1972). Perthite monzonites of similar composition, in other areas within the NS-block reveal much higher ages (1.73 Ga, Rb–Sr; 1.84 Ga, U–Pb; Skiöld, 1982) and the anomalous 1.53 Ga Rb–Sr age is regarded as an effect of disturbance of the radiometric system and not assigned to any geological event (Skiöld, 1987). The 1.84 Ga U–Pb age is therefore considered as a maximum age for these gabbro intrusions.

The gabbro-diabase (massif 3) and a layered intrusion, located further to the north (Tavinunnanen, massif 1), are regarded to define the end of major plutonic events in northern Sweden (Hallgren, 1979; Witschard, 1984).

In the southeastern part of the NS-block a mineralogically relatively well preserved gabbro (Kallax (4), Fig. 1) is sampled. This massif is surrounded by older gneisses and volcanic rocks. A few kilometers to the west of this gabbro there is a granite of 1.80 Ga in age (U–Pb, Öhlander and Schöberg, 1991). Pegmatites and aplites,

Table 2
Palaeomagnetic results from earlier studies within the Northern Svecofennian subprovince

Lat/Long. °N/°E	Locality		N/n	Decl.	Incl.	α_{95}	k	P_{lat}	P_{lon}	D_p/D_m	Magn.	Ref.
67.17/22.50	Tärendö	(13)	5/18	339	42	4.4	303	45	230	3/5	SF	1
67.67/21.13	Svappavaara	(14)	4/17	336	52	5.4	293	52	235	5/7	SF	1
67.67/21.67	Vittangi	(15)	8/28	340	39	5.4	107	43	228	4/6	SF	1
65.85/21.85	Notträsk	(16a)	5/22	346	35	3.6	460	43	220	2/4	SF	1
		(16b)	1/3	3	39	(4.4)	(792)	46	198	–	PSF	
67.83/20.58	Dundret	(17a)	4/15	193	3	7.7	143	21	187	4/7	SJ (A′)	1
		(17b)	1/5	15	13	(16.0)	(24)	28	184	–	SJ (A)	
		(17c)	1/5	175	−48	(8.2)	(88)	52	208	–	PSF	1
65.83/23.33	Sangis	(18)	11/35	359	40	2.8	264	47	205	2/3	PSF	1
65.90/24.10	Kukkola	(19a)	3/14	334	44	8.5	90	47	239	7/11	SF	2
		(19b)	1/3	174	10	(10.1)	(151)	17	208	–	SJ (A′)	
65.70/20.90	Korsträsk	(20)	1/10	5	47	(7.0)	(48)	53	194	–	PSF	3
68.22/21.95	Joulovaara	(21)	4/15	328	45	7.0	99	44	243	6/9	SF	2
66.00/22.10	Niemisel	(22)	4/24	340	35	8.8	110	41	227	6/10	SF	3
67.20/19.40	Stora	(23a)	3/18	343	33	16.3	58	40	221	11/19	SF	3
	Lulevattnet	(23b)	5/34	359	1	7.5	105	23	200	4/8	SJ (A)	
66.00/19.60	Jokkmokk	(24a)	2/11	349	33	–	–	41	214	–	SF	3
	basic rock	(24b)	3/3	11	−12	12.9	59	17	189	7/13	SJ (A′)	

For explanation see Table 1. References: 1 = Elming (1985); 2 = Elming (in prep.); 3 = Piper (1980).

probably related to the latest activity of the intrusion of this granite, penetrate the surrounding rocks and the gabbro. However, as the gabbro massif does not seem to be deformed by the intrusion of the granite, the two rocks may have intruded more or less contemporaneously.

The three southernmost gabbro massifs sampled are all related to the orogeny of the Skellefte district. Massif 7 (Näsberg, Fig. 1) is regarded to be contemporaneous with the oldest granites in the area, the Jörn granitoids (Kulling, 1955), which have ages of ~1.87 Ga (U–Pb, Wilson et al., 1987). Massif 6 (Bastutjärn) is considered to be related to the Revsund granitoids, which represent the latest geological event in this part of the shield (1.78 Ga, U–Pb, Skiöld, 1988). The well rounded, differentiated mafic intrusion 5 (Gallejaur, Enmark and Niska, 1984) is sampled in its more basic parts. The age of the massif of ~1.875 Ga (U–Pb, Skiöld, 1988; Skiöld et al., 1993), indicates that this rock was emplaced at about the same time as the Jörn granitoids.

The metasupracrustal rocks of the Skellefte district can be divided into two units: the marine Skellefte group and the younger, terrestrial, Arvidsjaur group. Radiometric dating and geological relations reveal an age of 1.877 Ga for the terrestrial volcanism (Skiöld et al., 1993). The Arvidsjaur volcanics were sampled in two sites (8 and 9, Fig. 1) and from the older Skellefte group samples from three sites (10, 11 and 12) were collected.

4. Palaeomagnetic results

The samples were collected with a portable drill and oriented using magnetic and sun compasses. The specimens were demagnetized by conventional alternating field (AF) and thermal demagnetization techniques and remanent magnetization measured with a spinner magnetometer (Molspin). The different components of remanence were defined by least square fit of vectors in two-dimensional plots (Torsvik, 1987).

The palaeomagnetic results obtained from massifs 1 and 2 (Table 1) remind a lot of those previously obtained from gabbro 17 (Table 2). At least two significantly different directions were

defined (Fig. 4a). These directions were identified in both massifs and the palaeomagnetic poles, A (1a: longitude = 199°E, latitude = 32°N and 2b: longitude = 193°E, latitude = 30°N), calculated from one of these groups of directions (1a: declination = 2°, inclination = 19°, 2b: declination = 6°, inclination = 15°) are very similar to the principal pole of massif 17 (17a: longitude = 187°E, latitude = 21°N, Fig. 8). These magnetizations are of both normal and reversed polarity (e.g. Fig. 2a). The remanence carrying the second group of directions, B (1b: declination = 24°, inclination = 46°, and 2a: declination = 14°, inclination = 51°) is when it occurs together with remanences carrying the direction A, isolated in a higher coercivity (Hc) and blocking temperature (Tb) range (Fig. 2b). This ultimate direction is best defined in the massif 2, where it more commonly occurs together with a soft magnetization (Fig. 2c) related to the present earth field (PEF).

The palaeomagnetic poles calculated from the directions B (1b: longitude = 169°E, latitude = 47°N, 2a: longitude = 180°E, latitude = 54°N) are quite different from previously determined poles from rocks in the NS-block (Fig. 8).

In some sites in massif 1 deviating directions of magnetization were traced (1c, Table 1), directions that may represent another generation (A′) of magnetization.

Apart from a very soft component carrying a PEF direction, only one component of magnetization was isolated in the gabbro-diabase, massif 3 (Fig. 3a). Previously identified directions of low inclination (Piper, 1980) were not obtained in this study. The mean direction of three sites (3: declination = 342°, inclination = 48°, Table 1) is very well defined and not significantly different from the direction of the SF massif 15 (Table 2, Figs. 4a and 5).

The mean direction of the characteristic magnetization of the massif 4 (4a: declination = 355°, inclination = 42°, Table 1) is significantly different from those of the massifs 1, 2 and 3 (Fig. 4a). If a soft magnetization carrying a PEF direction is not considered, this gabbro generally does not carry more than one component of magnetization. As indicated from the blocking temperatures the carrier of remanence is magnetite. However, at one site an anomalous direction is isolated (4b: declination = 21°, inclination = 57°, Table 1). This direction is poorly defined and not very different from the B directions of the massifs 1 and 2.

The mean directions of the remanent magnetizations of the Skellefte district gabbro massifs (5: declination = 344°, inclination = 35°, 6: declination = 341°, inclination = 41° and 7: declination = 343°, inclination = 48°, respectively, Table 1) are poorly defined and not significantly different from each other (Table 1, Fig. 4b). The characteristic magnetization in these massifs is generally of relatively high Tb (Figs. 3b and 3c) and Hc. Sometimes these rocks also carry large portions of soft remanence, which probably is of viscous origin, as the direction of magnetization is close to that of the PEF. The gabbro massif 6 is, contrary to the other massifs in this study, a rock with a low content of magnetite.

The volcanic rocks of the Skellefte district often carry two, sometimes poorly defined, components of magnetization. In the high Hc and Tb part of the remanence a direction of northerly declination and shallow inclination is defined (e.g. site 11, Table 1, Fig. 6). At lower Hc and Tb the remanence has a direction not significantly different from a SF direction (Figs. 6 and 4c) or that of PEF, which means that a viscous origin can not be excluded. The directions with low inclinations identified in the volcanic rocks at sites 9, 10 and 11 remind a lot of the A directions obtained in some of the gabbro massifs in the NS-block (massifs 1 and 2).

From the information of mineralogy (e.g. Ambros and Henkel, 1973; Larson et al., 1984; O. Martinsson, pers. commun., 1993) and blocking temperatures, the carriers of remanences in the gabbro massifs are suggested to be magnetite–titanomagnetite.

In the volcanic rocks and in one gabbro (massif 6) of the Skellefte district pyrrhotite sometimes occurs. In an earlier study of magnetic properties of pyrrhotite from the Skellefte district (Elming, 1982) the remanent magnetization of this mineral is shown to be very soft and

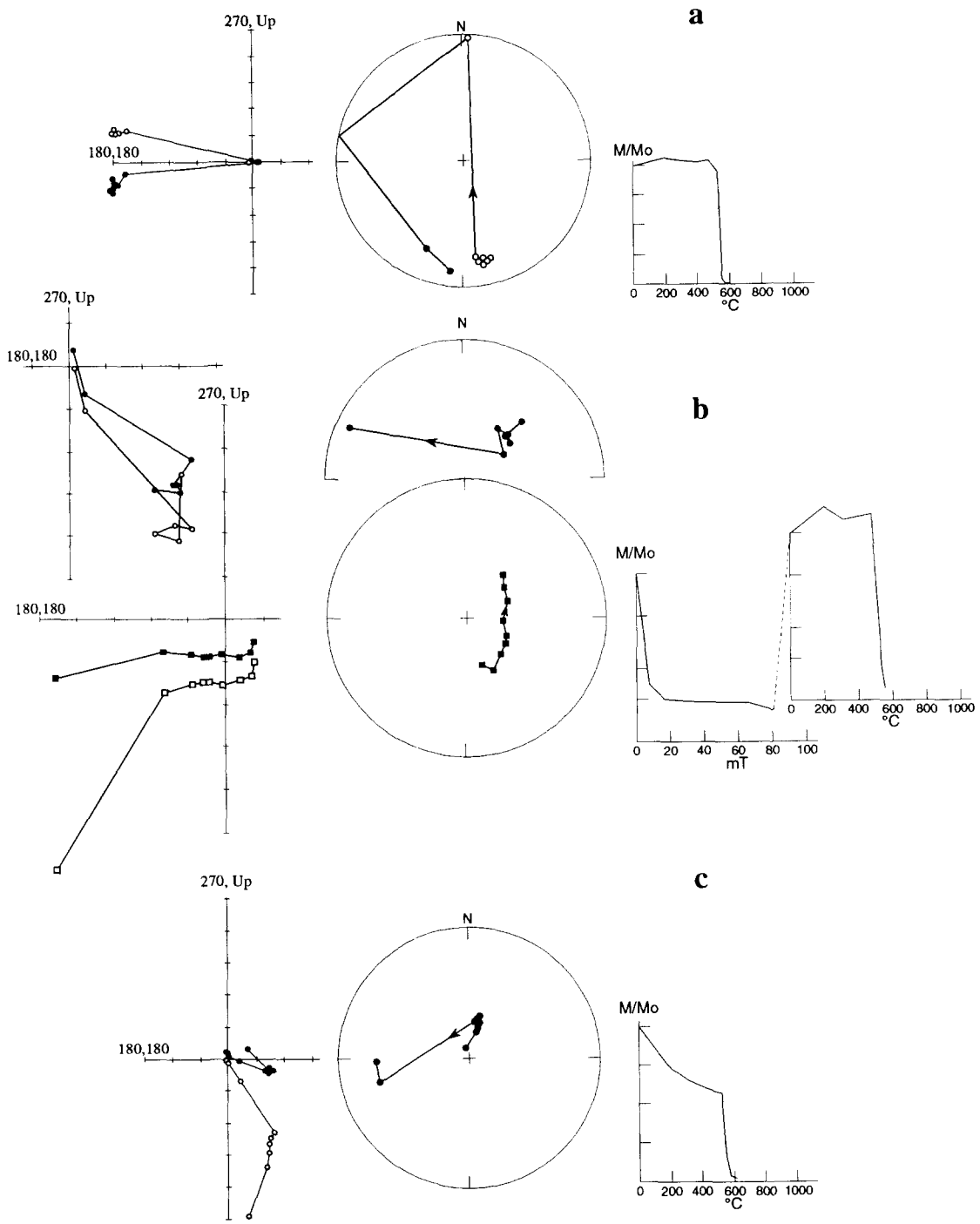


Fig. 2. Stereographic and vector plots showing examples of behaviour of remanent magnetization during alternating field (AF) and thermal demagnetization of specimens from the Tavinunnanen gabbro (a and b) and the Akkavaare gabbro (c); (b) shows a specimen demagnetized first by AF and then by thermal technique.

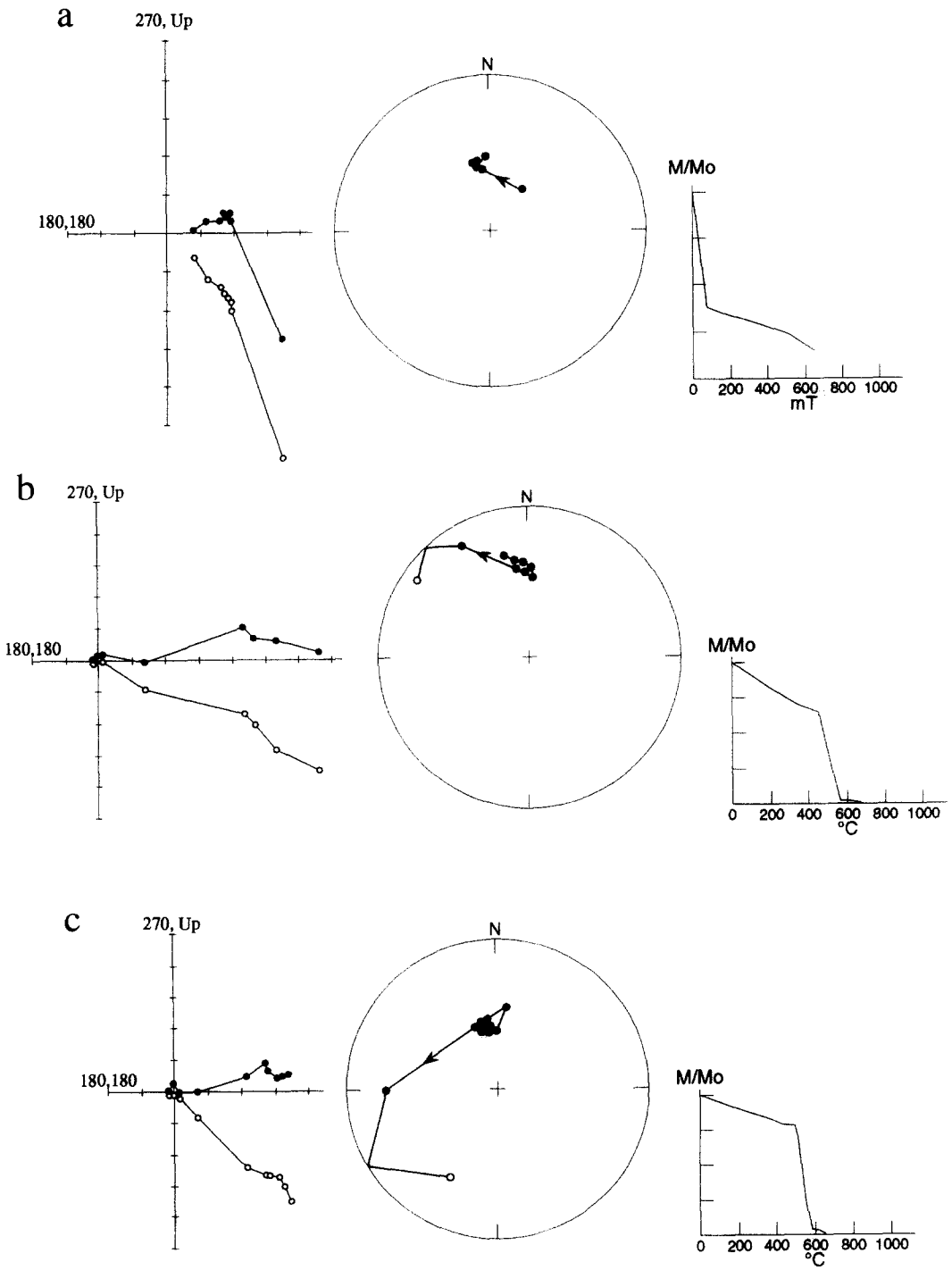


Fig. 3. Examples of remanence behaviour during alternating field and thermal demagnetization shown in stereographic and vector plots. The specimens are from: (a) the Nabbrejarka (3), (b) the Gallejaur (5), (c) the Bastutjärn (6) gabbro massifs.

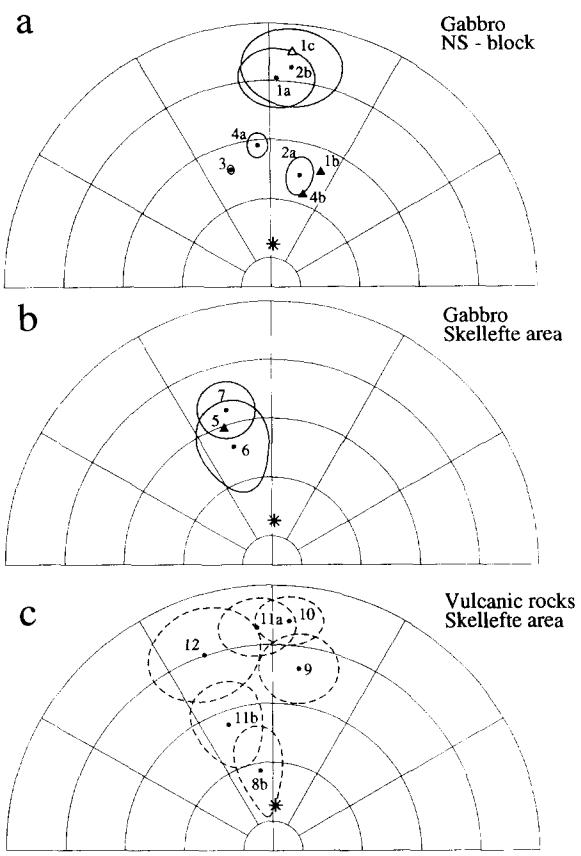


Fig. 4. Mean directions and confidence cones of the remanent magnetizations identified in the various gabbro massifs and volcanic rocks of this study. The direction of the present earth field is indicated with a star. Mean directions calculated from less than three sites are plotted as triangles.

reflecting the direction of the PEF. The high Hc and Tb in these rocks is considered to be carried by magnetite–titanomagnetite.

5. Tectonics

The degree of tectonic influence on the gabbro massifs varies. Some are affected only to a very small extent (e.g. massifs 15 and 16; Elming, 1982, 1985; Widenfalk et al., 1985), while others have been subject to more intense faulting (e.g. massifs 13 and 14, Elming, 1982, 1985).

The tectonic history of the Skellefte district is different from that of the NS-block. The volcanic rocks have been subject to two generations of folding and the latest is related to the intrusion of the Revsund granites and gabbros. The volcanic rocks were during this event folded with a shallow dipping WNW-trending fold axis (Grip, 1945).

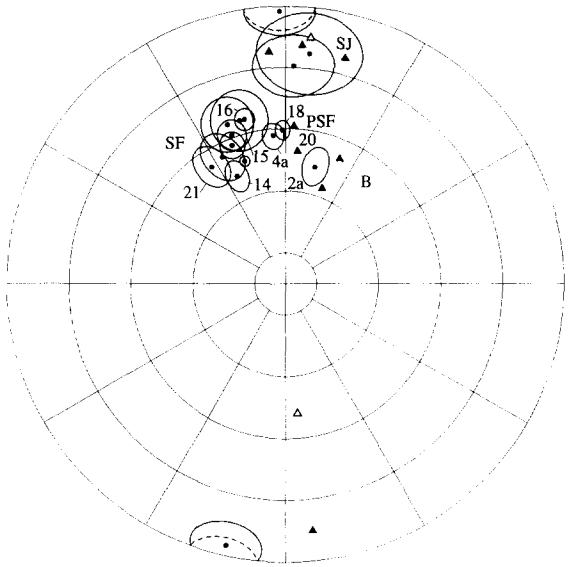


Fig. 5. Mean directions and confidence cones of magnetizations identified in gabbros and volcanic rocks of this study and previous studies in the Northern Svecofennian subprovince and the Skellefte district. Four groups of directions can be recognized, denoted as *SF* (Svecofennian), *PSF* (post-Svecofennian), *B* (Late post-Svecofennian) and *SJ* (Subjotnian). Mean directions calculated from less than three sites are plotted as triangles.

The rocks of the Skellefte group were sampled where they trend E–W and dip ca. 40° to the south (site 12, Fig. 1). A tectonic correction of the poorly defined direction of magnetization at site 12 (Table 1) results in a direction (declination = 326°, inclination = 56°) not very different from those of the older gabbro massifs in the NS-block (e.g. massifs 14 and 15, Table 2). This may be reasonable as the rocks of the Skellefte group are older than 1.877 Ga. However, the volcanic rocks are usually altered showing signs of metamorphic grades ranging from low greenschist to amphibolite facies (Claesson, 1986). This is reflected in the magnetization of the volcanic rocks at sites 10 and 11. The rocks are here folded, but still the site mean directions

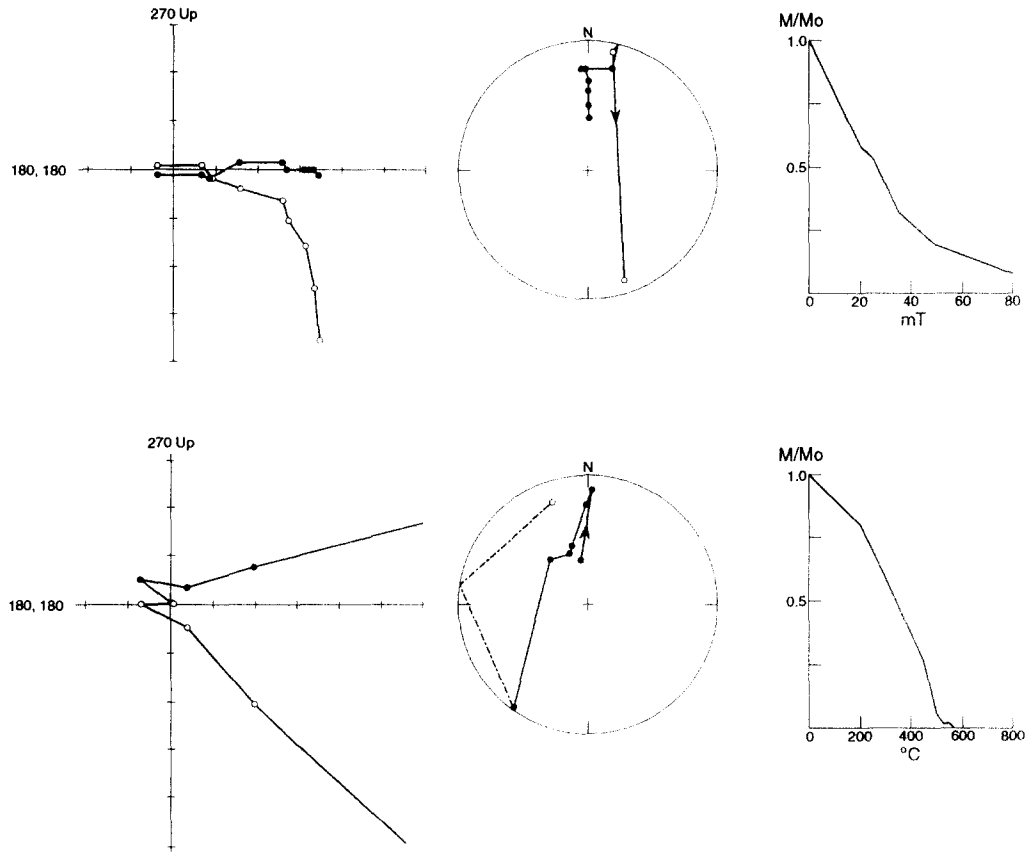


Fig. 6. Behaviour of remanent magnetization during AF and thermal demagnetization of specimens of a volcanic rock (site 11) from the Skellefte district.

10 and 11a are not significantly different from each other or the mean direction of site 12 (Fig. 4c). This means the rocks are probably remagnetized. The low-coercivity component at site 11 (11b) may also be a post-tectonic SF magnetization, as before any correction it coincides with the SF directions (e.g. 15, Fig. 5).

The gabbro massif 6, presumably related to the intrusion of the Revsund granites, seems not to be tectonically affected (Gavelin, 1955). Also the older gabbro massifs within the Skellefte district are rather well preserved; however, faults of normal and strike-slip origin are observed.

In the NS-block the gabbro-diabase (3, Fig. 1) is faulted all along its extension. These faults are of a strike-slip type and the directions of characteristic magnetization of sites from different parts of the gabbro are not significantly different.

The massif 2 is cut by strike-slip and normal faults into a northern and southern part. Horizontal structures are found in the gabbro from the northern part (Ambros and Henkel, 1973), indicating that this part, from which samples of this study were collected, has probably not been tilted.

From the shape, structure and mineral chemistry, Larson et al. (1984) suggested that the layered gabbro 1 was faulted and slightly tilted (15–20°). A correction for tilt does not significantly change the directions of remanences; however, the direction 1b after correction (declination = 38°, inclination = 34°) is more different from what has been defined as B directions in other massifs (Tables 1 and 2).

Also in the gabbro massif 4 layering is observed. This layering trends north–south and dips 50° to the east. Assuming the layering was originally horizontal, a correction for tilt results in a mean direction for the massif (4a corrected: declination = 33°, inclination = 29°) quite similar to the corrected 1b direction. However, considering the remanence direction 1b is poorly defined, this is not a good example of a positive tectonic test. The uncorrected direction 4a is not significantly different from the mean remanence direction of massif 18 (Table 2, Fig. 5), which is also located in the southeastern part of the NS-block (Fig. 1).

Based on the discussion above, tectonic corrections might be rewarding only for some of the gabbro massifs and in the further discussion the directions considered (Tables 1 and 2) are all tectonically uncorrected.

6. Radiometric data and ages of magnetization

6.1. Radiometric ages

U–Pb dating of zircons has successfully been used for dating granitic rocks in northern Sweden (Skiöld, 1987). Plotting the age data in a diagram they show three different frequency maxima (Fig. 7), one of which reflects the SF orogeny (1.89–1.86 Ga). Archaean–Palaeoproterozoic ages are predominantly found in the northern part of the NS-block, but determined also in rocks from isolated Archaean cores in the southeast (Fig. 7). The frequency maximum in the interval 1.80–1.76 Ga reflects a post-SF generation of granites. Rocks representing this generation are found almost all over the investigated area, but with a concentration to the southwest. This may be apparent, only reflecting the uneven distribution of sampled localities.

Rb–Sr age determinations, predominantly whole-rock, have been made on rocks from the same areas as those of the U–Pb datings. Sometimes samples have been used both for U–Pb and Rb–Sr age determinations. In these cases the Rb–Sr ages are generally lower than the U–Pb ages. The age distribution of the Rb–Sr data (Fig. 7) is also different from that of the U–Pb, showing only one well defined frequency maximum. The ages related to this peak (1.75–1.72 Ga) may possibly correspond to the 1.80–1.76 Ga U–Pb ages, reflecting a lower closing temperature for the Rb–Sr system. The Rb–Sr ages in this interval are like the post-SF U–Pb age data concentrated in the southern region.

Between 1.65 Ga and 1.50 Ga there are a number of ages which do not show any peak in the distribution. The Rb–Sr whole-rock systems corresponding to these ages are not regarded to represent geochronologically correct systems and the obtained ages therefore have no geochronological significance (Romer, 1994).

Rb–Sr data indicating younger rocks are reported preferentially from the Caledonian area.

K–Ar age data are geographically rather irregularly distributed and most of the data are from the southern part of the area investigated (Fig. 7). These data do not show any specific peak in the age distribution. Ages from 2.015 Ga to 1.15 Ga are reported, but with a slight concentration to the interval 1.65–1.60 Ga. Ages within this interval are preferentially found in the southwestern part of the area, where also post-Svecofennian (PSF) U–Pb ages are determined. Slow uplift and cooling and the K–Ar system closing at lower temperatures may imply that these ages reflect the same post-SF event as was indicated by the U–Pb data.

As mentioned above, Pb isotope analyses indicate a Caledonian ($\sim$0.43 Ga) reactivation of the Proterozoic basement outside the Caledonides.

6.2. Ages of magnetization

Due to the difficulty in finding zircons in gabbroic rocks for U–Pb age determinations, only a few of the gabbro massifs have been radiometrically dated. This means that ages of magnetization most often have to be estimated from the geological relation of the rocks to the dated rocks.

Granitoids in the marginal parts of massif 15 (Vittangi, Table 2) were dated at 1.89 Ga (Skiöld et al., 1988). Sm–Nd isotope analyses of the granitoids and the gabbro massif indicate that

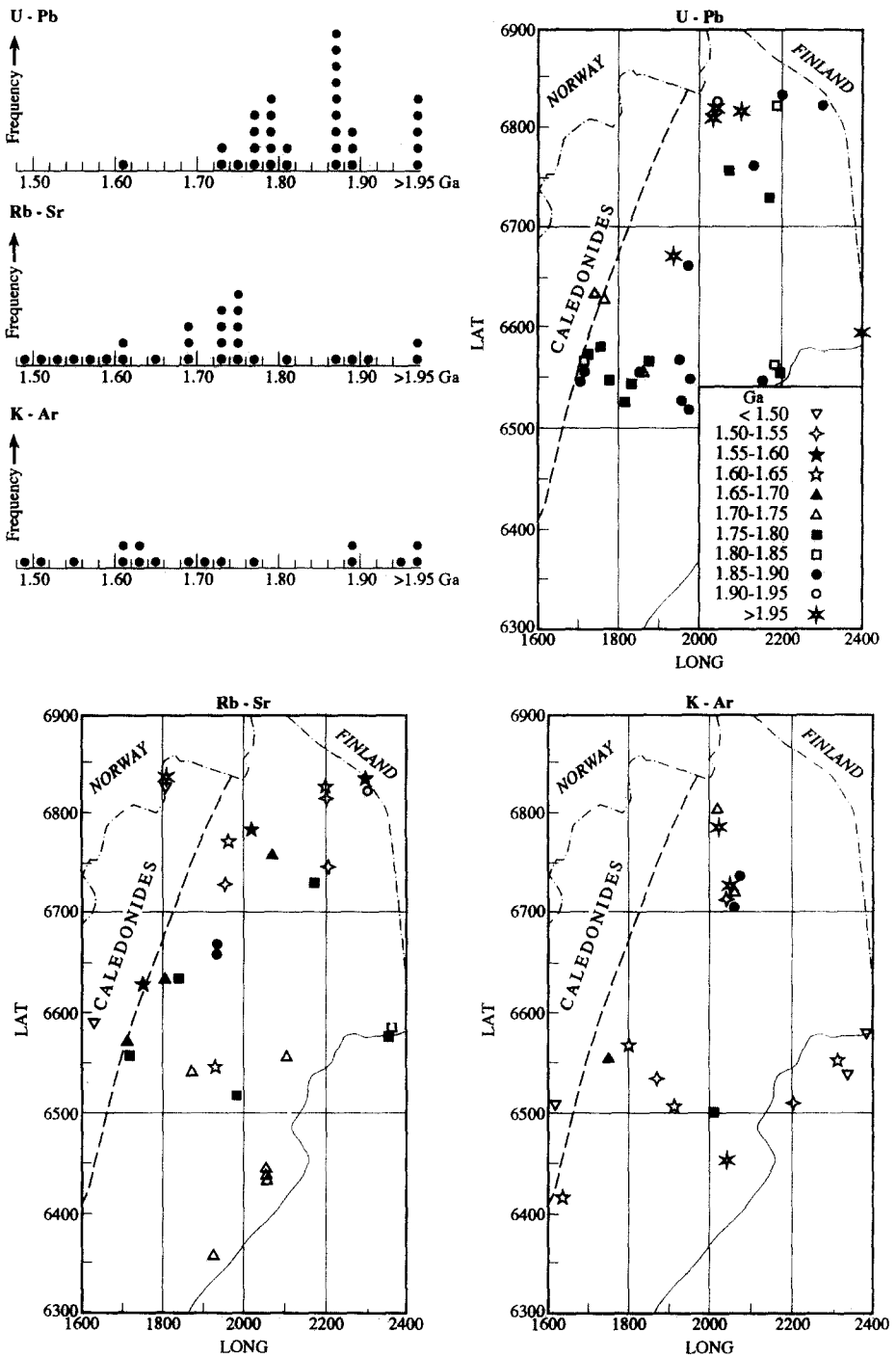


Fig. 7. Frequency distribution and geographical distribution of age data referring to U–Pb, Rb–Sr and K–Ar age determinations. The age data are partly from Wilson and Sundin (1979).

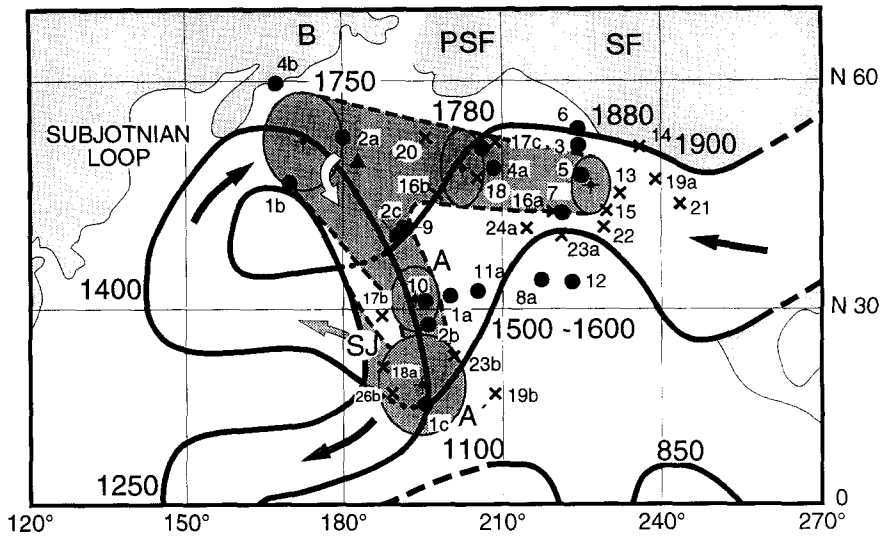


Fig. 8. The apparent polar wander path of Pesonen et al. (1989) and suggested changes based on virtual geomagnetic poles (VGP's) of this study. The codes used for the VGP's are the same as those in Fig. 1. The dot denotes data from this study; the dot with plus-sign denotes the grand means accompanied by A_{95}, the half-angle of the 95% circle of confidence. The triangle denotes a possible Late Svecofennian pole reported by Mertanen et al. (1989), and × denotes poles determined in previous studies within the Northern Svecofennian subprovince.

they are genetically related and may therefore be regarded as contemporaneous. Also gabbro 14 (Svappavaara, Table 2) is believed to be of the same generation; however, its irregular shape may imply that it is somewhat older than the tectonically unaffected gabbro 15. In Elming (1985) the mean palaeomagnetic pole of massif 14 was also regarded to represent a magnetization older than that of massif 15, which here serves as a reference for a magnetization at 1.89 Ga.

A very well defined pole is calculated from the mean remanence direction of the gabbro-diabase (3, Table 1). Unfortunately there are no radiometric data in relation to this rock. However, the gabbro intrudes rocks, which are supposed to be genetically related to 1.84 Ga old granites and the age of these granites may therefore be regarded as a maximum age of the magnetization. The pole position (3, Fig. 8) also correlates with a SF age of magnetization.

Except for the marginal parts of massif 16 (Notträsk, Fig. 1), the gabbroic rocks from northern Sweden of this and earlier studies (Elming, 1985) generally show few signs of alteration. The altered part of massif 16 carries a secondary magnetization (Elming, 1985) with a calculated pole (16b, Table 2, Fig. 8) significantly different from the mean pole of the massif, but close to that of massif 18, which is also located in the southeastern part of the NS-block. Massif 16 has been radiometrically dated at a poorly defined age of 1.8 Ga (Sm–Nd, P.J. Hamilton, pers. commun., 1985).

The pole of massif 4 (4a, Table 1, Fig. 8) falls close to that of the secondary pole of massif 16 and the mean pole of massif 18. This gabbro may be contemporaneous with the nearby Ale granite of 1.80 Ga (Öhlander and Schöberg, 1991) and this age is also regarded as a probable magnetization age. The palaeomagnetic pole is significantly different from the SF reference pole 15 and together with the poles 16b and 18 it is considered as a post-SF magnetization.

From massif 5, dated at 1.876 Ga (Skiöld et al., 1993), only two sites may be used for the calculation of a mean pole. A comparison of this pole (5, Table 1) with the SF reference pole 15 (1.89 Ga) shows that the poles are very close to each other (Fig. 8), which means the magnetization is probably original.

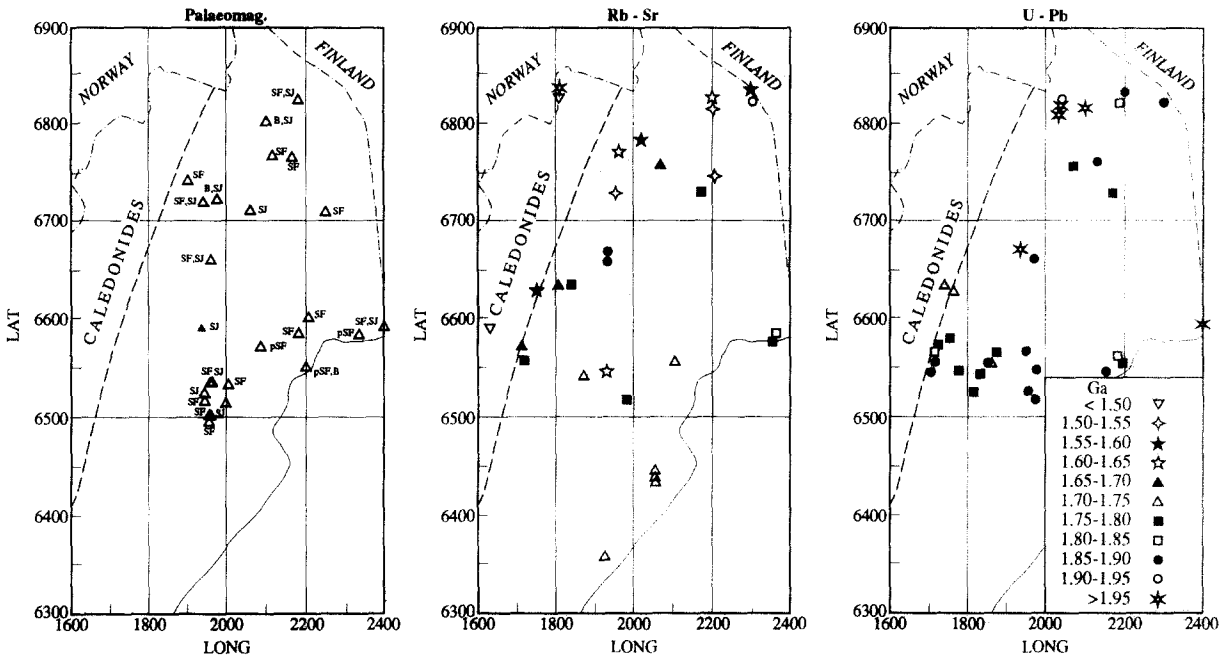


Fig. 9. The geographical distribution of rocks carrying the different generations of magnetization. *SF*=Svecofennian age, *PSF*=post-Svecofennian, *B*=Late PSF, *SJ*=Subjotnian. For a comparison with the geographical distributions of the different generations of magnetization the corresponding distribution of Rb–Sr and U–Pb age data are plotted.

Table 3
Grand mean palaeomagnetic poles and reference directions from 1.88 Ga to ~1.50 Ga for the Northern Svecofennian subprovince and the Skellefte district

Magnetization	Age	Massif	*N*	*D*-ref	*I*-ref	Lat-ref	P_{lat}	P_{lon}	A_{95}
SF	1880	3,5,6,7,13,14 15,16a,19a,21,24a	12	346	38	21	46	227	3.5
PSF	1780	4a,16b,17c,18,20	5	5	39	22	48	201	4.0
PSF (B)	1750	4b,2a,1b	3	20	51	32	53	171	7.6
SJ (A)	1500–1600	1a,2b,9,10,11a, 17b,23b	7	11	11	6	31	195	4.9
SJ (A′)	1500–1600	1c,17a,19b,24b	4	12	−14	−7	18	195	8.0

Note. Magnetization=symbols for the different generations of magnetization. Age=age of magnetization. Massif=symbols for the mean directions of the massifs and sites used for the grand mean calculations. *N*=number of mean directions used for calculation of the grand mean, *D*-ref, *I*-ref=reference declination and inclination calculated from the grand means with respect to the reference location (Kajaani, 64.1°N, 27.7°E). Lat-ref.=reference palaeolatitude. P_{lat} and P_{lon}=latitude (°E) and longitude (°N) of the grand mean pole positions (see Fig. 4), A_{95}=half-angle of 95% circle of confidence of the grand mean poles.

The magnetization ages of massifs 6 and 7 can be estimated only from their relations to the Revsund and Jörn granites, respectively. The ages of these granites, 1.78 Ga and 1.87 Ga, respectively, indicate that significantly different remanence directions may be expected. However, the mean remanence directions and poles are poorly defined (Table 1, Fig. 4b) and not significantly different. When comparing them with the SF and post-SF directions and poles it can only be concluded that the magnetizations can be of both SF and post-SF origin.

The directions of magnetization and calculated poles from massifs 1 and 2 are partially different from those of the other massifs in this study. The corresponding magnetization ages are also more difficult to define. Pole positions similar to the 1a and 2b poles have been calculated from the characteristic magnetization of massif 17 (Fig. 8) and as secondary magnetizations in other gabbro massifs. In the altered part of the gabbro massif 21 (Fig. 1), a secondary magnetization carrying an A direction has been traced (Elming, in prep.). This gabbro and surrounding granodiorites have been radiometrically dated. The Rb–Sr ages are low, 1.62 Ga and 1.5–1.7 Ga, respectively (Skiöld, 1979), while the U–Pb age of the granodiorite is SF (~1.86 Ga, Skiöld, 1982). The nature of this A direction as being a secondary magnetization is also confirmed from a study of basic dykes in northern Sweden (Elming and Öhlander, in prep.). In that study, the A direction is isolated as a secondary magnetization only in dykes where geochemical analyses indicate a significant degree of alteration.

In southern Finland and central Sweden, SJ dykes (e.g. Fölgö dykes, 1.57 Ga, U–Pb, Suominen, 1991) carry magnetizations with directions very similar to the A directions (Pesonen et al., 1989). In the NS-block a SJ U–Pb age has been recorded only in one locality (Skiöld, 1986).

Palaeomagnetic poles of Caledonian age have been reported from other parts of the Fennoscandian shield. These poles (e.g. longitude = 344°E and latitude = 19°S, longitude = 352°E and latitude = 19°S; Douglass, 1988, and Trench and Torsvik, 1991, respectively) are not very different, but of opposite polarity, from the A poles (e.g. 1a: longitude = 199°E, latitude = 32°N). This means a Caledonian origin of the A magnetization would be possible. However, the Caledonian reactivation of the Proterozoic basement is considered to have caused only low-grade metamorphism defined in mylonite zones and often bound to supracrustal rocks (Romer, 1990). Moreover, the A poles are sometimes calculated from rocks sampled more than 100 km east of the Caledonian front (Fig. 9) and they are therefore not considered to represent Caledonian magnetizations.

These poles fall on a part of the APWP where the path overlaps itself (Fig. 8) and may on this basis correspond to a magnetization age of both ~1.50 Ga and 1.30 Ga. No radiometric age of ~1.30 Ga has been reported from northern Sweden, while Early Subjotnian ages have been traced (Skiöld, 1986). An magnetization age of 1.50 to 1.60 Ga is therefore assigned to the direction A.

The SJ (A) magnetization is in the massifs 1 and 2 isolated together with what is here defined as the B direction of magnetization (1b, decl. = 24°, incl. = 46°, 2a, decl. = 14°, incl. = 51°). The B component is isolated in a Tb and Hc range higher than that of the SJ component. Mineralogical studies of magnetite from massif 1 indicate the magnetite to be of primary origin, with very few signs of alteration (S.Å. Larsson, pers. commun., 1992). This suggests that the B magnetization has not a chemical origin and should be regarded as a magnetization older than the SJ magnetization. A Late Svecofennian age (~1.75 Ga) was suggested for a corresponding direction of magnetization in a layered intrusion in northern Finland (Mertanen et al., 1989). Geological information from northern Sweden suggests the intrusion of massifs 1 and 2 to define the end of plutonic activities in the area and thereby they mark the end of the post-SF episode (~1.80–1.75 Ga). An age of ~1.75 Ga is therefore suggested for the mean palaeomagnetic pole B (Table 3, Fig. 8).

The poles calculated from the magnetization of volcanic rocks of the Skellefte and Arvidsjaur groups in the Skellefte district do not indicate any significant trend in ages. The volcanic rocks are strongly tectonized and metamorphosed and poles correlating with SJ and SF magnetizations are therefore regarded as being the result of remagnetization. Based on the palaeomagnetic data it is therefore not possible to define a more specific age or tectonic signature to these different rockunits.

7. Grand mean palaeomagnetic poles

On the basis of the palaeomagnetic data presented here no significant tectonic difference between the Skellefte district and the NS-block can be defined. SF or SJ directions of magnetization determined in rock from both blocks are not significantly different (Figs. 4 and 5) and Grand mean poles are therefore calculated using data from both tectonic regimes (Table 3).

Radiometric data indicate the SF gabbro massifs to have intruded within a period of ~30 Ma. There is a trend in the distribution of the mean directions from relatively high positive inclinations and northwesterly declinations to more shallow inclinations and north-northwesterly declinations. This is demonstrated by the mean directions of e.g. massifs 14 and 21, which are significantly different from those of massifs 15 and 16 (Fig. 5). Combined with data from other parts of the shield Early, Middle and Late Svecofennian directions can also be defined (Pesonen et al., 1989; Elming et al., 1993).

However, the lack of radiometric data from the gabbro massifs in north Sweden makes such a subdivision of SF directions difficult and the SF is here only represented by one grand mean (Table 3).

The B poles, which probably represent a magnetization older than the SJ magnetization, are also calculated from overprints in Svecofennian (SF) rocks. These poles are together with the post-SF poles considered to be related to the period of intrusion of the post-SF (PSF) granitoids.

Looking at the mean directions of magnetization, which are based on a minimum of three sites from each massif, two means corresponding to the PSF poles (4a and 18) and one to the B pole (2a) can be defined. The PSF mean directions are significantly different from all but one of the SF mean directions and different from the B mean direction (Figs. 4 and 5). Including means of individual sites the picture is not that clear. Site means 4b and 1b are significantly different from the PSF mean directions 4a and 18, while site mean 20 (Table 2, Fig. 5) is not significantly different from either PSF or B directions. On the basis of the best defined mean directions 4a, 18 and 2a (Fig. 5), the PSF and the B magnetizations are regarded to represent two different generations of magnetization. Both generations may be related to the time period 1.80–1.75 Ga. With reference to the above discussion the PSF magnetization age is ~1.80 Ga and the B magnetization probably ~1.75 Ga. As the PSF magnetizations are determined in rocks predominantly from the southern part of the area, the pole of massif 20 (see Fig. 1) is pooled together with the PSF group for the calculation of a grand mean pole (Table 3).

The grand mean poles corresponding to the A directions is calculated from poles defined in the massifs 1, 2, 17 and 24 and in site means of the volcanic rocks of the Skellefte district (Table 3). Two groups of directions are identified. One is of negative and low inclination (A′) and another of positive and somewhat higher inclination (A). Both groups have declinations close to the north–south. One site mean of the volcanic rocks (9) is excluded from the grand means as it has a direction of magnetization not significantly different from either PSF or A directions. The mean directions, used for the calculation of the poles, are all significantly different from the other groups of directions (Fig. 5).

8. Block tectonics

So far, on the basis of palaeomagnetic data, no intraplate tectonics within the Fennoscandian shield has been demonstrated (Pesonen et al., 1989; Elming et al., 1993). This is further substantiated by the comparison of Svecofennian poles from the Skellefte district and the Northern Svecofennian subprovince.

Poles from these blocks are not significantly different from poles of similar ages in other blocks of the Fennoscandian shield (see Pesonen et al., 1989) and the drift history of northern Sweden may therefore be representative for the whole shield.

The drift history presented here is partly based on poor data and should therefore be regarded as tentative. The orientation and latitudinal position of the Skellefte and the Northern Svecofen-

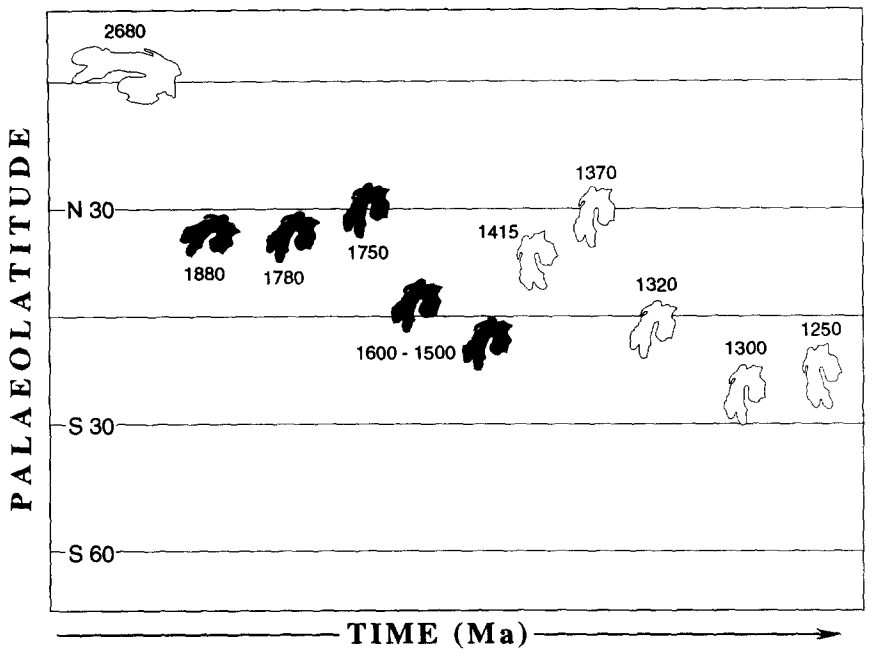


Fig. 10. The drift history of Fennoscandia from 1.88 Ga to 1.50 Ga based on palaeomagnetic data from the Northern Svecofennian subprovince and the Skellefte district. Fennoscandia is plotted at correct palaeolatitudes and in correct orientations, while it has been shifted arbitrarily to the right as its longitudinal position cannot be determined by palaeomagnetic methods. The black figures show the shield positions calculated in this study, while the white ones are from Pesonen et al. (1989). The variation in size of Fennoscandia is due to the projection (Gall's).

nian subprovince (NS-block) during 1.88–1.50 Ga were calculated from the grand mean poles (GMPs). The results (Fig. 10) demonstrate that these blocks and the Fennoscandian shield from Svecofennian (~1.89 Ga) to post-SF (1.80 Ga) time drifted northward and meanwhile rotated in an anticlockwise sense (ca. 19°). This drift pattern continued until the end of the post-SF period (~1.75 Ga). An abrupt change in the latitudinal drift direction took place from the end of PSF to Subjotnian. Fennoscandia now drifted with a higher velocity (ca. 0.3°/Ma) to a position close to the Equator. This drift history suggests a modification of that presented by Pesonen et al. (1989). In the present model, Fennoscandia continued its northward movement until ~1750 Ga, while in the former Fennoscandia started to move southward already at ~1.80 Ga.

9. Conclusion

Four different directions of remanent magnetizations have been isolated in gabbro massifs in northern Sweden.

(1) A Svecofennian magnetization has been referred to an age in the interval 1.89–1.86 Ga. The magnetization (SF) is carried by rocks from all over the northern part of Sweden. This emphasizes the strong influence of this orogeny on the originally Archaean basement of the Northern Svecofennian subprovince.

(2) A second magnetization (PSF) is interpreted to reflect a post-Svecofennian (1.80 Ga) geological event, which from the geographical distribution of the radiometric data seems most frequent in the southern part of the area investigated. Basic intrusions carrying PSF directions of magnetization are also preferentially found in the southeast.

(3) A possible late PSF (~ 1.75 Ga) origin has been ascribed to the B directions of magnetization. This magnetization may reflect a generation of layered intrusions that define the end of the post-SF geological event. This means that there have been two, maybe three different generations of gabbro intrusions in northern Sweden.

(4) Subjotnian directions of magnetization (A) are identified as overprints in gabbroic and volcanic rocks. Subjotnian magmatism is described from central Sweden (Porter, 1975; Piper, 1979; Bylund, 1985) and southern Finland (Johnson, 1979; Suominen, 1991), and the palaeomagnetic results of this study show that magmatic activity of this age also affected the northwestern part of the Fennoscandian shield.

The magnetization of volcanic rocks of the Skellefte district are found to be post-tectonic. Subjotnian and possibly Svecofennian remagnetizations are identified.

Baesed on these palaeomagnetic data no large-scale relative movements since Svecofennian time can be indicated between the Skellefte district and the Northern Svecofennian subprovince north of the Skellefte district.

The results of this study added new data to the apparent polar wander path of Fennoscandia and tentatively modified the drift history for the shield as presented by Pesonen et al. (1989) for the period 1.88–1.50 Ga.

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